

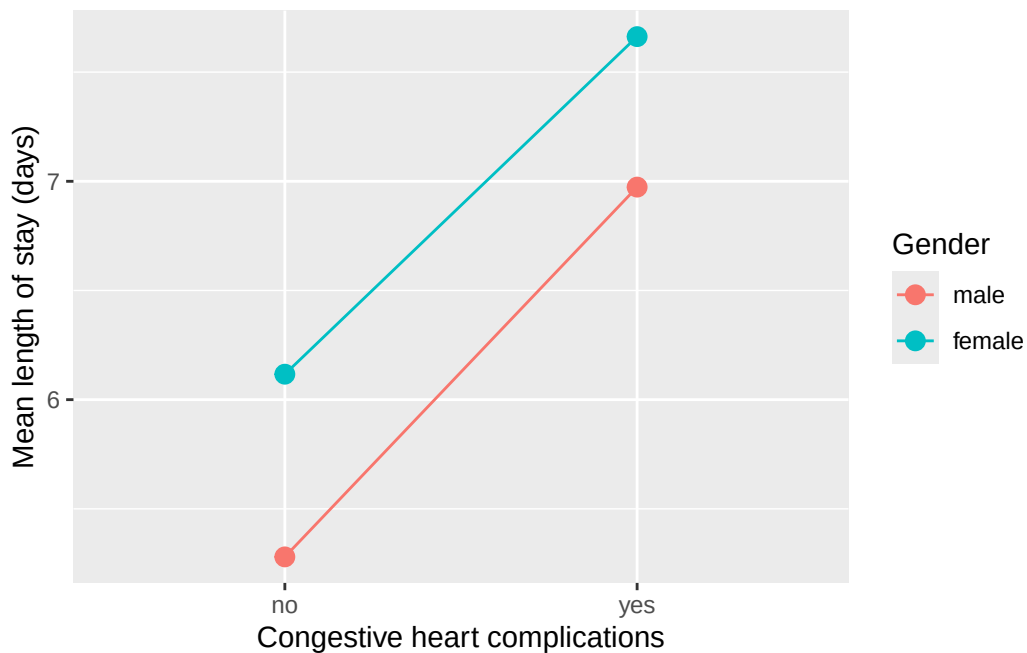
Medical Statistics – Answers lab 8

Part 1: Estimated marginal means in a two-way ANOVA

Load the data

Exploratory plot

```
whas500 |>
  group_by(gender, chf) |>
  summarise(
    mean_los = mean(los, na.rm = TRUE),
    .groups = "drop"
  ) |>
  ggplot(aes(x = chf, y = mean_los, group = gender, color = gender)) +
  geom_point(size = 3) +
  geom_line() +
  labs(
    x = "Congestive heart complications",
    y = "Mean length of stay (days)",
    color = "Gender"
  )
```



Question: Based on the interaction plot, do the lines look approximately parallel, or is there visual evidence that the association between CHF and length of stay differs by gender?

Answer:

The lines look approximately parallel. Patients with CHF have a longer mean length of stay for both males and females, and the size of this difference appears similar across gender. There is no strong visual evidence of an interaction.

Two-way ANOVA model

```
fit_emm <- lm(los ~ gender * chf, data = whas500)
anova(fit_emm)
```

Analysis of Variance Table

Response: los

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
gender	1	127.7	127.720	5.9285	0.0152495 *
chf	1	275.4	275.427	12.7847	0.0003837 ***
gender:chf	1	0.6	0.563	0.0262	0.8715880
Residuals	496	10685.6	21.543		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Question: Inspect the ANOVA table. Is there evidence of an interaction between **gender** and **chf**?

Answer:

No. The interaction between **gender** and **chf** is not statistically significant, $F(1, 496) = 0.03$, $p = 0.872$. We therefore do not focus on formal simple-effect contrasts. Instead, we interpret the main effects and their estimated marginal means.

Estimated marginal means and contrasts

```
emmeans(fit_emm, ~ gender)
```

gender	emmean	SE	df	lower.CL	upper.CL
male	6.13	0.309	496	5.52	6.73
female	6.89	0.335	496	6.23	7.55

Results are averaged over the levels of: chf
Confidence level used: 0.95

```
emmeans(fit_emm, ~ chf)
```

chf	emmean	SE	df	lower.CL	upper.CL
no	5.70	0.262	496	5.18	6.21
yes	7.32	0.373	496	6.59	8.05

Results are averaged over the levels of: gender
Confidence level used: 0.95

```
emmeans(fit_emm, ~ gender * chf)
```

gender	chf	emmean	SE	df	lower.CL	upper.CL
male	no	5.28	0.309	496	4.67	5.89
female	no	6.12	0.424	496	5.28	6.95
male	yes	6.97	0.536	496	5.92	8.03
female	yes	7.66	0.519	496	6.64	8.68

Confidence level used: 0.95

```
# Contrast tests for the main effects
pairs(emmeans(fit_emm, ~ gender))
```

contrast	estimate	SE	df	t.ratio	p.value
male - female	-0.763	0.456	496	-1.673	0.0950

Results are averaged over the levels of: chf

```
pairs(emmeans(fit_emm, ~ chf))
```

contrast	estimate	SE	df	t.ratio	p.value
no - yes	-1.62	0.456	496	-3.552	0.0004

Results are averaged over the levels of: gender

```
# Contrast tests for simple effects
# Only focus on these if the interaction is meaningful/statistically significant.
pairs(emmeans(fit_emm, ~ chf | gender))
```

gender = male:

contrast	estimate	SE	df	t.ratio	p.value
no - yes	-1.69	0.619	496	-2.736	0.0064

gender = female:

contrast	estimate	SE	df	t.ratio	p.value
no - yes	-1.55	0.670	496	-2.307	0.0214

```
pairs(emmeans(fit_emm, ~ gender | chf))
```

chf = no:

contrast	estimate	SE	df	t.ratio	p.value
male - female	-0.837	0.525	496	-1.595	0.1114

chf = yes:

contrast	estimate	SE	df	t.ratio	p.value
male - female	-0.689	0.746	496	-0.924	0.3560

Question: What is the estimated mean length of stay for males and females, averaged over CHF status?

Answer:

Averaged over CHF status, the estimated marginal mean length of stay is approximately **6.13 days** for males and **6.89 days** for females. The main-effect contrast is not statistically significant at the 0.05 level ($p = 0.095$).

Question: What is the estimated mean length of stay for patients with and without CHF, averaged over gender?

Answer:

Averaged over gender, the estimated marginal mean length of stay is approximately **5.70 days** for patients without CHF and **7.32 days** for patients with CHF. The main-effect contrast is statistically significant ($p < 0.001$), indicating that patients with CHF have a longer estimated mean length of stay.

Question: Should we focus on main-effect contrasts or simple-effect contrasts?

Answer:

Because the interaction is not statistically significant, we focus on the **main-effect contrasts**. The simple-effect contrasts are shown to demonstrate how they can be requested, but they are not the primary basis for interpretation in this example.

Part 2: Building prediction models using backward elimination

Step 1: Fit the initial linear regression model

Create an initial model for hospital length of stay (`los`) using the following predictors: `age`, `gender`, `hr`, `sysbp`, `diasbp`, `bmi`, `cvd`, `sho`. Run/summarize the model to inspect coefficients and p-values.

Call:

```
lm(formula = los ~ age + gender + hr + sysbp + diasbp + bmi +  
    cvd + sho, data = whas500)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.335	-2.653	-1.071	1.200	40.073

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.2648144	2.1394896	1.526	0.127659

```

age          0.0031994  0.0168850  0.189 0.849792
genderfemale 0.8575246  0.4489334  1.910 0.056698 .
hr           0.0190577  0.0090828  2.098 0.036396 *
sysbp       -0.0008358  0.0085656  -0.098 0.922304
diasbp      0.0141799  0.0128884  1.100 0.271780
bmi         -0.0304532  0.0427613  -0.712 0.476700
cvdyes      0.3903925  0.4981468  0.784 0.433600
shoyes      3.5265170  1.0329265  3.414 0.000693 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.632 on 491 degrees of freedom
Multiple R-squared:  0.04991,    Adjusted R-squared:  0.03443
F-statistic: 3.224 on 8 and 491 DF,  p-value: 0.001388

```

Step 2: Eliminate the least significant predictor

Anova Table (Type III tests)

```

Response: los
      Sum Sq  Df F value    Pr(>F)
(Intercept)  50.0   1  2.3286 0.1276592
age           0.8   1  0.0359 0.8497924
gender       78.3   1  3.6486 0.0566978 .
hr          94.5   1  4.4025 0.0363962 *
sysbp        0.2   1  0.0095 0.9223042
diasbp      26.0   1  1.2105 0.2717799
bmi         10.9   1  0.5072 0.4766997
cvd         13.2   1  0.6142 0.4336001
sho        250.1   1 11.6561 0.0006929 ***
Residuals   10535.8 491
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

The ANOVA table shows that the predictor with the highest p-value is **sysbp** ($p = 0.92$). Systolic blood pressure is the least significant predictor and should be removed from the model.

Step 3: Repeat the steps

The following variables are sequentially removed from the model (after initially removing sysbp):

1. age: p-value = 0.86
2. cvd: p-value = 0.41
3. bmi: p-value = 0.44
4. diasbp: p-value = 0.22

Step 4: Final model

```
final_model <- lm(los ~ gender + hr + sho, data = whas500)
summary(final_model)
```

Call:

```
lm(formula = los ~ gender + hr + sho, data = whas500)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.663	-2.661	-1.064	1.136	40.826

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.796075	0.799047	4.751	2.66e-06	***
genderfemale	0.910388	0.424827	2.143	0.032602	*
hr	0.020680	0.008843	2.339	0.019751	*
shoyes	3.550448	1.009081	3.518	0.000474	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.622 on 496 degrees of freedom

Multiple R-squared: 0.04443, Adjusted R-squared: 0.03865

F-statistic: 7.687 on 3 and 496 DF, p-value: 4.976e-05

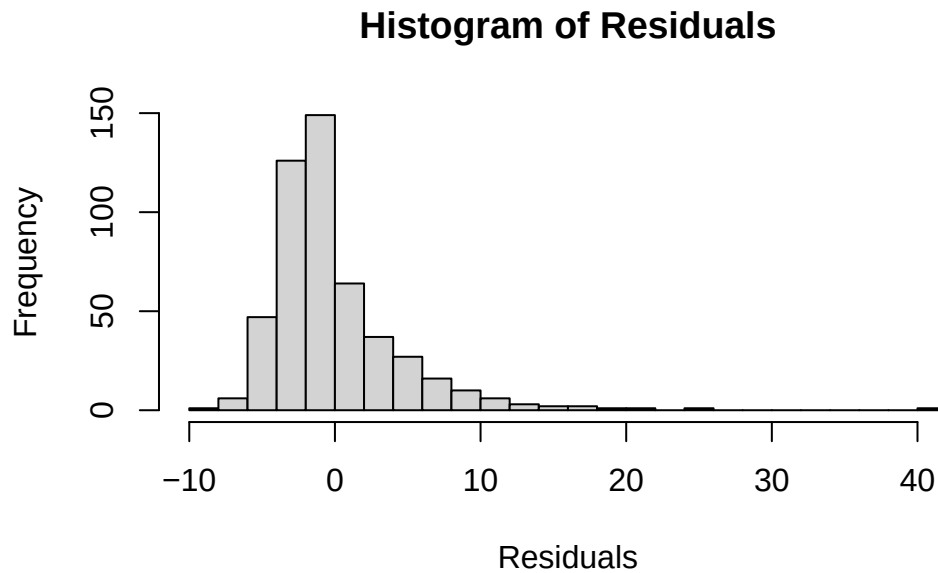
```
# 95% CIs for regression coefficients
confint(final_model)
```

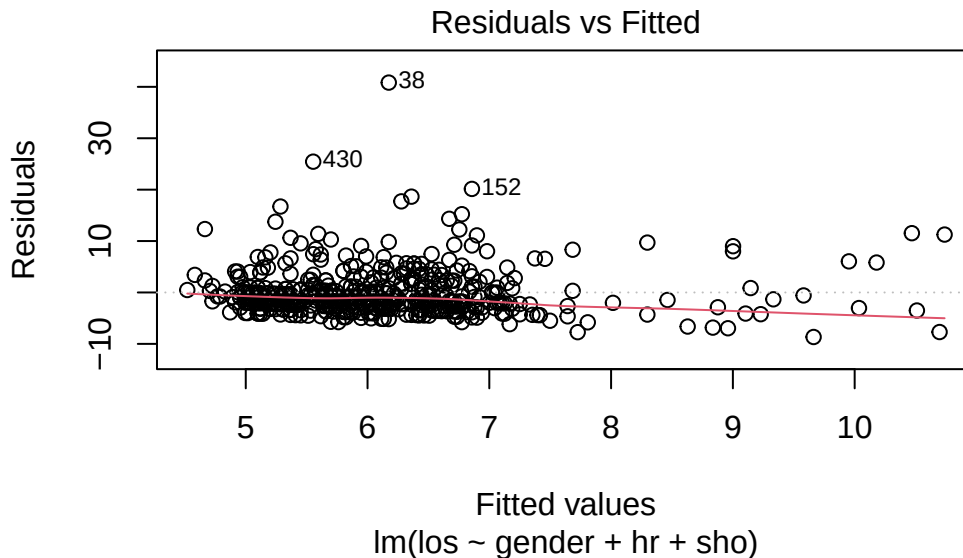
	2.5 %	97.5 %
(Intercept)	2.226140295	5.36601012
genderfemale	0.075705855	1.74507053
hr	0.003306032	0.03805437
shoyes	1.567847029	5.53304829

The final model for hospital length of stay includes **gender**, **hr**, and **sho**. All predictors have $p < 0.10$.

Predictor	Coefficient	95% CI	p-value
(Intercept)	3.80	[2.23, 5.37]	< 0.001
Gender (Female)	0.91	[0.08, 1.75]	0.033
Heart rate	0.02	[0.003, 0.038]	0.020
Cardiogenic shock	3.55	[1.57, 5.53]	< 0.001

Residual plots





The overall fit of the model appears reasonable, as the residuals are generally centered around zero with no major patterns suggesting severe violations of linearity. However, there are some outliers with very long lengths of stay (LOS) that are not adequately captured by the model. These outliers lead to a right skew in the residual distribution, as seen in the histogram, influencing model fit. While the current model seems to work reasonably well for most observations, further steps (e.g., transformations or robust regression techniques) could be considered to better account for these extreme cases.

Part 3: Automated procedures for building prediction models (logistic regression)

In this part, we explore automated procedures for predictor selection in **logistic regression** prediction models, focusing on predicting in-hospital death (`dstat`).

R

```
# Create a 0/1 outcome (1 = dead)
whas500 <- whas500 |>
  mutate(dstat01 = ifelse(dstat == "dead", 1, 0))

fit_full <- glm(
  dstat01 ~ age + gender + hr + sysbp + diasbp + bmi + cvd + sho,
  family = binomial,
```

```

data = whas500
)

fit_step <- stepAIC(fit_full, direction = "backward", trace = FALSE)
summary(fit_step)

```

Call:

```

glm(formula = dstat01 ~ age + hr + sysbp + sho, family = binomial,
    data = whas500)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-6.186164	1.687679	-3.665	0.000247	***
age	0.059272	0.016933	3.500	0.000464	***
hr	0.013961	0.007497	1.862	0.062572	.
sysbp	-0.017333	0.006212	-2.790	0.005271	**
shoyes	3.061710	0.525719	5.824	5.75e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 273.86 on 499 degrees of freedom
Residual deviance: 204.09 on 495 degrees of freedom
AIC: 214.09

Number of Fisher Scoring iterations: 6

```

# Odds ratios and 95% CIs
exp(cbind(OR = coef(fit_step), confint.default(fit_step)))

```

	OR	2.5 %	97.5 %
(Intercept)	0.002057705	7.530576e-05	0.05622609
age	1.061063592	1.026428e+00	1.09686811
hr	1.014058710	9.992675e-01	1.02906886
sysbp	0.982816468	9.709221e-01	0.99485659
shoyes	21.364054186	7.624145e+00	59.86543899

Question: Which predictors are retained in the prediction model?

Answer:

The final logistic regression model obtained via backward elimination using AIC includes the following predictors: **age**, **hr** (heart rate), **sysbp** (systolic blood pressure), and **sho** (cardiogenic shock).

SPSS

In SPSS, using the **Backward: LR** method (standard settings), the procedure also yields a final model with these four predictors.

Model if Term Removed				
Variable	Model Log Likelihood	Change in -2 Log Likelihood	df	Sig. of the Change
Step 1				
age at hospital admission	-103,678	6,032	1	,014
initial heart rate	-102,521	3,719	1	,054
initial systolic blood pressure	-101,967	2,610	1	,106
initial diastolic blood pressure	-101,029	,734	1	,392
bmi	-101,172	1,021	1	,312
gender	-101,121	,920	1	,338
cardiogenic shock	-117,707	34,091	1	<,001
history of cardiovascular disease	-100,665	,006	1	,940
Step 2				
age at hospital admission	-103,786	6,244	1	,012
initial heart rate	-102,522	3,714	1	,054
initial systolic blood pressure	-101,970	2,611	1	,106
initial diastolic blood pressure	-101,031	,733	1	,392
bmi	-101,178	1,028	1	,311
gender	-101,132	,934	1	,334
cardiogenic shock	-117,812	34,296	1	<,001
Step 3				
age at hospital admission	-105,218	8,373	1	,004
initial heart rate	-102,603	3,145	1	,076
initial systolic blood pressure	-105,209	8,356	1	,004
bmi	-101,531	1,000	1	,317
gender	-101,513	,963	1	,326
cardiogenic shock	-117,818	33,573	1	<,001
Step 4				
age at hospital admission	-106,522	10,019	1	,002
initial heart rate	-103,164	3,302	1	,069
initial systolic blood pressure	-105,576	8,126	1	,004
bmi	-102,043	1,062	1	,303
cardiogenic shock	-118,275	33,524	1	<,001
Step 5				
age at hospital admission	-109,385	14,682	1	<,001
initial heart rate	-103,733	3,380	1	,066
initial systolic blood pressure	-106,252	8,417	1	,004
cardiogenic shock	-118,724	33,362	1	<,001

Figure 1: SPSS results of the backward selection for logistic regression

Part 4: Causal diagrams

For each of the exercises below:

- First solve the diagram by hand using the recipe from the lab.
- Pay special attention to the meaning of a path: a path connects variables, but does not have to follow the direction of the arrows.
- Then check your answer using the [DAGitty webtool](#).

Exercise 1

In the graph depicted below, for which variables do you need to adjust to assess the unconfounded effect of E on O (there may be several possibilities)?

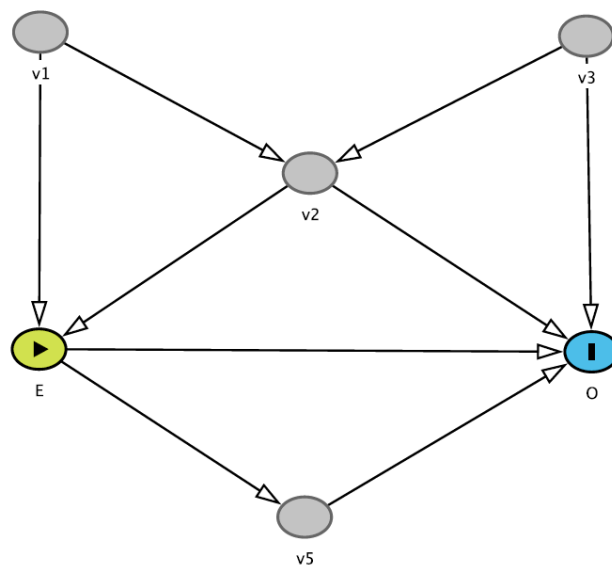


Figure 2: DAG exercise 1

Answer:

Using the recipe, there are several non-causal paths between E and O that need to be considered. Adjusting for v2 blocks one path, but it also opens a path through a collider structure (E - v1 - v3 - O). This newly opened path needs to be closed by also conditioning on v1 or v3, or both. Hence, there are 3 options: (v1, v2, v3) ; (v1, v2) ; and finally, (v2, v3).

Exercise 2

In the graph depicted below, what happens when you additionally adjust for **v5**?

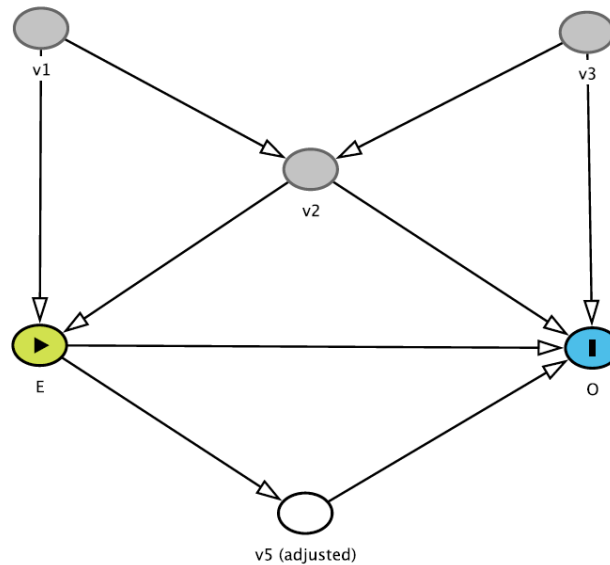


Figure 3: DAG exercise 2

Answer: When adjusting for **v5**, we are blocking the effect through this indirect path from **E** to **O** (**v5** is a mediator between **E** and **O**). Instead of the total effect of **E** on **O**, we will be estimating the direct effect.

In DAGitty, when you set **v5** to 'adjusted', the algorithm will say the following: "The total effect cannot be estimated due to adjustment for an intermediate or a descendant of an intermediate."

Exercise 3

This diagram is slightly different: **v1** now is the exposure. For which variables do you need to adjust to assess the unconfounded effect of **v1** on **O**?

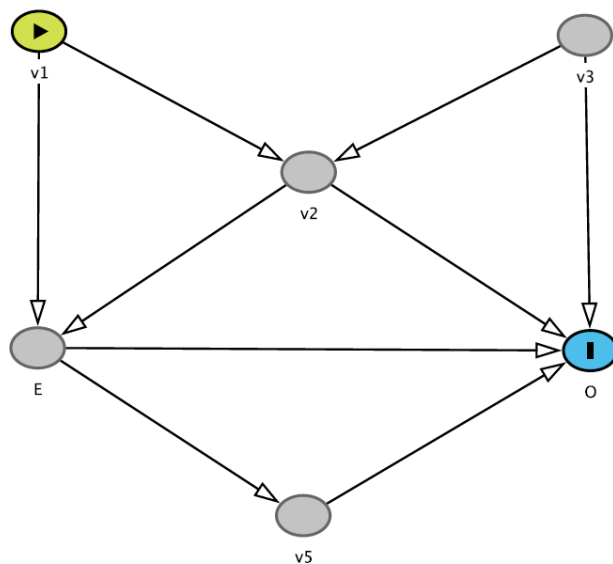


Figure 4: DAG exercise 3

Answer: No adjustment is needed: there are no backdoor paths (removing all arrows leaving $v1$ reveals no remaining unblocked path from $v1$ to O).

Exercise 4

Now, $v2$ is the exposure. For which variables do you need to adjust to assess the total unconfounded effect of $v2$ on O ?

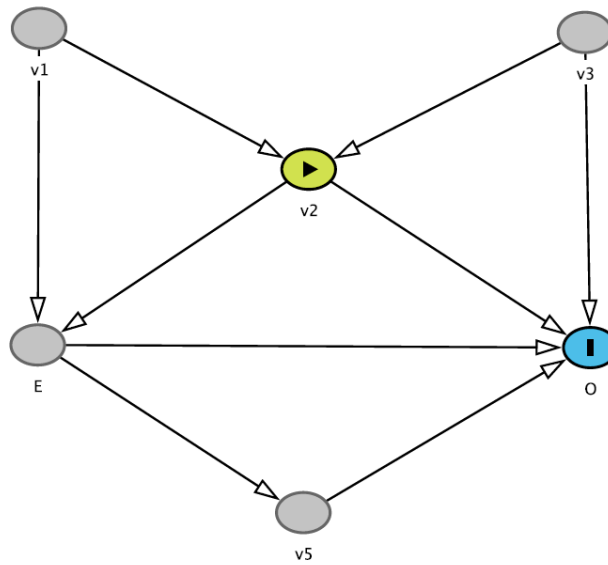


Figure 5: DAG exercise 4

Answer: Using the recipe, there are three non-causal paths that need to be blocked when $v2$ is the exposure:

- a) $v2 - v3 - O$ and
- b) $v2 - v1 - E - O$
- c) $v2 - v1 - E - v5 - O$

Backdoor path a) can be closed by conditioning on $v3$.

Backdoor path b) can be closed by conditioning on $v1$ (but not by conditioning on E , as you would no longer be estimating the total effect by blocking the paths from $v2$ to O mediated by E).

In this case, you should therefore condition on $v1$ and $v3$.

Exercise 5

Back to the first DAG. However, $v2$ is now unmeasured. Can we still obtain an unconfounded estimate of the effect of E on O ?

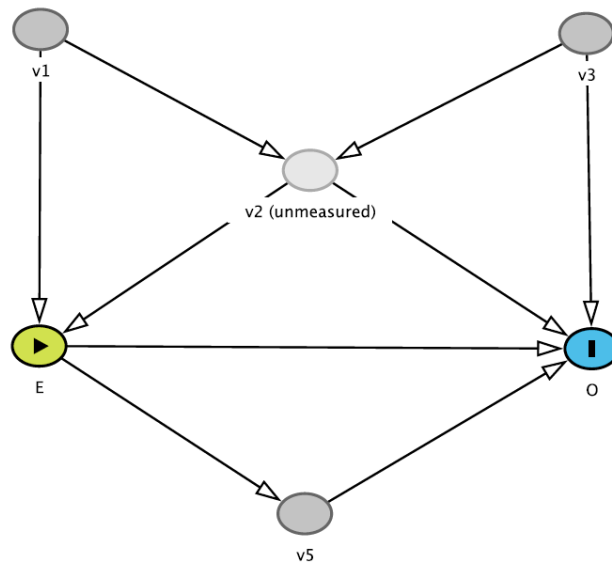


Figure 6: DAG exercise 5

Answer: No, we cannot close the backdoor path between E and O since v2 is unmeasured and cannot be corrected for.

Exercise 6

See the DAG below: you adjusted for **v5**. What would be the consequence of this action?

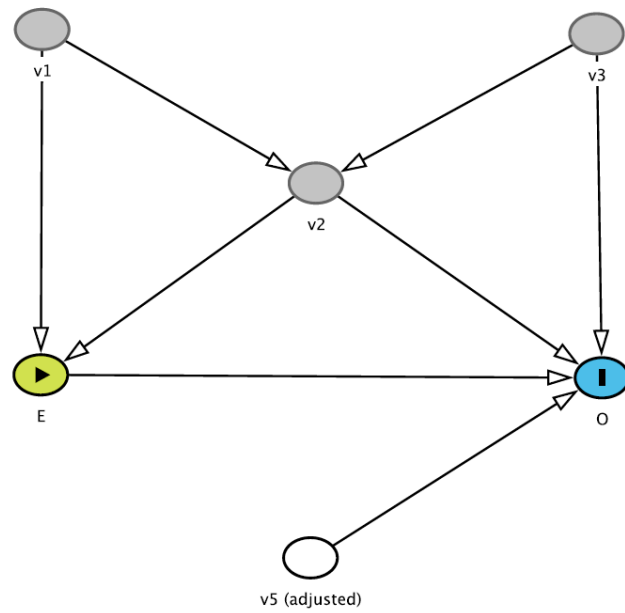


Figure 7: DAG exercise 6

Answer: There is no consequence: conditioning on v_5 cannot alter any of the estimated effects in the DAG (it is neither a confounder, collider, nor a mediator in the E-O relationship).